

About Me

- Started as a Data Engineer
- A few years in a medical imaging start-up working on lung cancer detection in x-rays, and brain tumours in CT scans
- Approaching 4 years at Visia, building Computer Vision applications for recycling and heavy industry.

Visia

- Started off selling cameras to recycling facilities
- Chucked x-rays in for good measure (and preventing fires from Lithium Ion Batteries)
- Added Lidar into the mix for some applications
- Now Multisensor AI for Heavy Industry

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What we Do

Where do we use Agents

Types of Context

GEPA Algorithm

GEPA Context

Context Pressure

Thank You

What we Do

Help with cost
disagreements
between buyers
and sellers at
metal yards.



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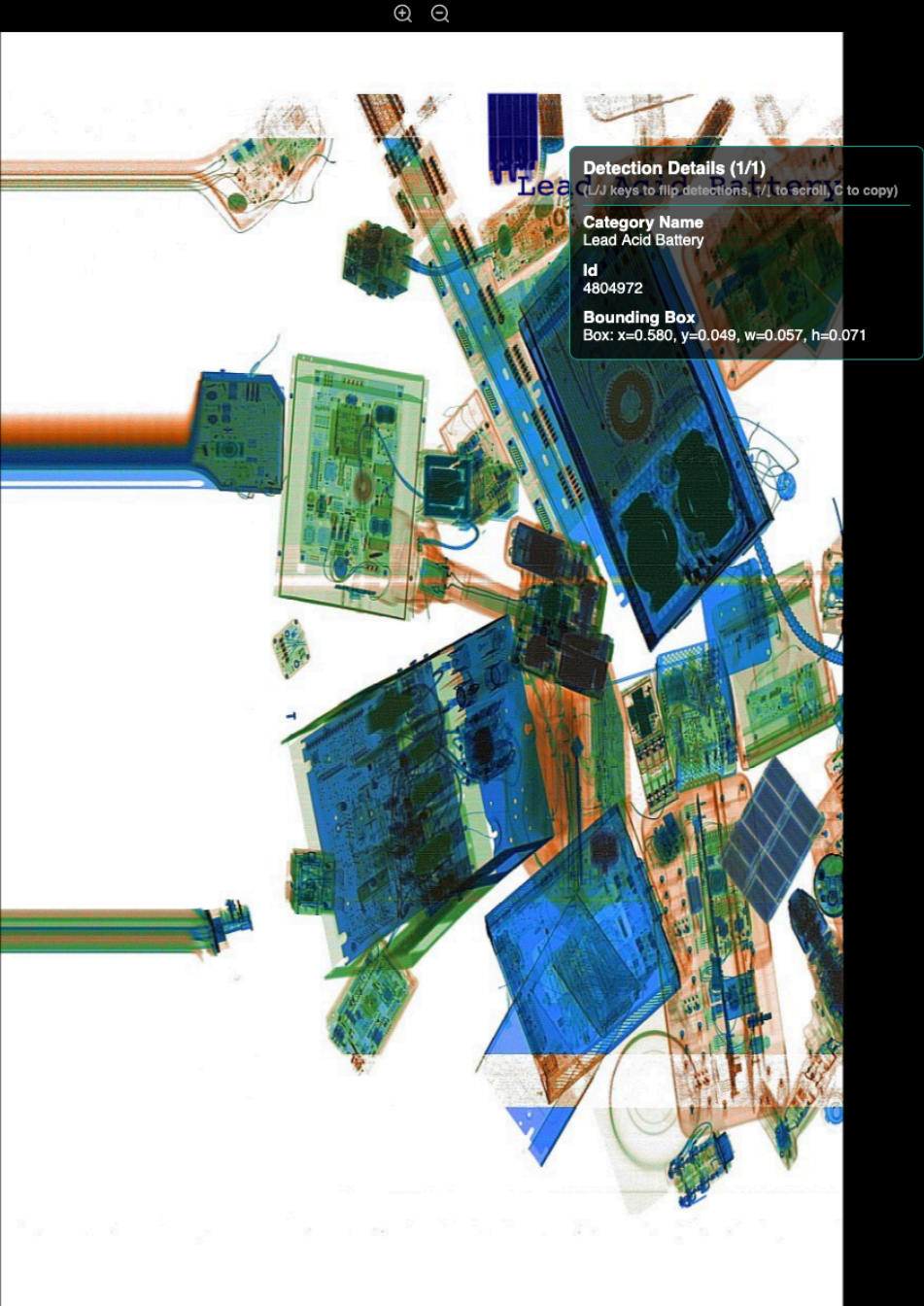
GEPA Context

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What we Do

Find batteries in e-waste and municipals recycling with x-rays and lasers.



What we Do

Detect and send notifications for 'bulkies' in waste to energy facilities.



Where do we use Agents

- **Billing disagreements in metal yards**
- **Active learning simulations**

Types of Context

- **Prompt context** — system instructions and framing baked into the initial call
- **Retrieved context (RAG)** — information pulled from external stores at query time
- **Conversational context** — the history of the current interaction
- **Tool/observation context** — results from tool calls and API responses the agent generates for itself
- **Visual context** — composite images encoding predictions, ground truth, or crops for a downstream model
- **Persistent context** — memories and session state carried across conversations (the cookies of agents, but more like a cheatsheet intentionally kept small)

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GEPA Algorithm

GEPA's core algorithm iterates through three stages — **Executor**, **Reflector**, **Curator** — each with distinct context demands.

Stage	Role	Context
Executor	Runs candidate program on a small training minibatch (<code>reflection_minibatch_size</code> , default 3 examples)	Captures full execution traces: reasoning chains, intermediate outputs, tool calls, error messages
Reflector	Feeds traces + evaluator feedback into a strong LLM (<code>reflection_lm</code>)	Diagnoses failure modes and identifies causal patterns
Curator	Proposes concrete instruction mutation based on the diagnosis	Transforms reflection into actionable prompt edits

GEPA Context Management

GEPA (Generalized Evolutionary Prompt Adaptation) manages context through:

- **Evaluation cache** — stores `(candidate, example)` results to avoid redundant inference
- **Reflective dataset** — captures execution traces for reflection-based prompt mutation
- **Pareto frontiers** — tracks best programs per validation example or objective (compressed historical context)
- **Batch sampling** — strategic minibatch selection balances coverage vs. cost
- **State persistence** — `GEPASState` serializes candidate evolution.

Handling Context Pressure

Practical mitigations for the double-pressure problem:

- **Reduce** `reflection_minibatch_size` from 3 to 1–2 for ReAct programs with long trajectories
- **Use a high-context reflection LM** — models with large context windows (10M tokens ideal)
- **Reduce ReAct's** `max_iters` to 3–5 instead of 20
- **Keep tool return values concise** — control retrieved passage counts and output verbosity
- **Override** `truncate_trajectory()` for domain-aware truncation that preserves the most informative steps, not just the most recent

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Be kind to each other, AI's getting wild.

We're Hiring

If this kind of work interests you, reach out.